

AIML Complete Syllabus (Basic to Advanced)

1. Python Programming (Beginner to Intermediate)

- Syntax, Variables, Data Types
- Conditionals and Loops
- Functions, Modules, File Handling
- Data Structures: Lists, Tuples, Sets, Dictionaries
- OOP Concepts: Classes, Objects, Inheritance
- Libraries: NumPy, Pandas, Matplotlib, Seaborn

2. Mathematics for AIML

- Linear Algebra: Vectors, Matrices, Eigenvalues
- Probability & Statistics: Bayes' Theorem, Distributions
- Calculus: Derivatives, Gradients (for backpropagation)
- Optimization: Gradient Descent

3. Data Handling & Visualization

- Data Cleaning & Preprocessing
- Handling Missing Values, Outliers
- Exploratory Data Analysis (EDA)
- Visualizing data with Matplotlib & Seaborn

4. Machine Learning (ML) Fundamentals

- Supervised Learning: Linear/Logistic Regression, SVM, KNN
- Unsupervised Learning: KMeans, PCA, Hierarchical Clustering
- Model Evaluation: Accuracy, Precision, Recall, F1, ROC
- Overfitting/Underfitting, Cross-Validation
- Feature Engineering & Selection

5. Deep Learning (DL)

- Neural Networks: Perceptrons, Activation Functions
- Forward & Backpropagation

AIML Complete Syllabus (Basic to Advanced)

- CNNs: For Image Processing
- RNNs/LSTMs: For Time-Series and NLP
- Frameworks: TensorFlow, Keras, PyTorch

6. Natural Language Processing (NLP)

- Text Preprocessing: Tokenization, Stop Words, Stemming
- Bag of Words, TF-IDF, Word Embeddings (Word2Vec, GloVe)
- NLP Models: RNNs, LSTMs, Transformers (BERT, GPT)
- Applications: Sentiment Analysis, Chatbots, Translation

7. Computer Vision

- Image Processing Basics (OpenCV, PIL)
- CNNs for image classification
- Object Detection (YOLO, SSD)
- Image Segmentation, Face Recognition

8. Reinforcement Learning

- Markov Decision Process (MDP)
- Q-Learning, SARSA
- Deep Q-Networks (DQN)
- Applications: Game AI, Robotics

9. Tools & Platforms

- Jupyter Notebooks, Google Colab
- Git/GitHub for version control
- ML Pipelines with Scikit-Learn, MLFlow
- Deployment: Flask/FastAPI, Streamlit, Heroku

10. Capstone Projects & Portfolio

- End-to-End ML Projects (Data -> Model -> Deployment)

AIML Complete Syllabus (Basic to Advanced)

- Kaggle Competitions
- Portfolio Website with GitHub Projects
- Resume Building for AIML roles